

Figure 1. One platform, two modes and a new AI detection layer (live energy OT)

SOC Mode = deception-based active defence. Research Mode = offline analyst copilot. A new AI detection layer (ML anomaly detection + attacker-behaviour classification, developed in this project) sits between them, trained on real traffic and live decoy data.

Energy operator OT environment (Purdue model)

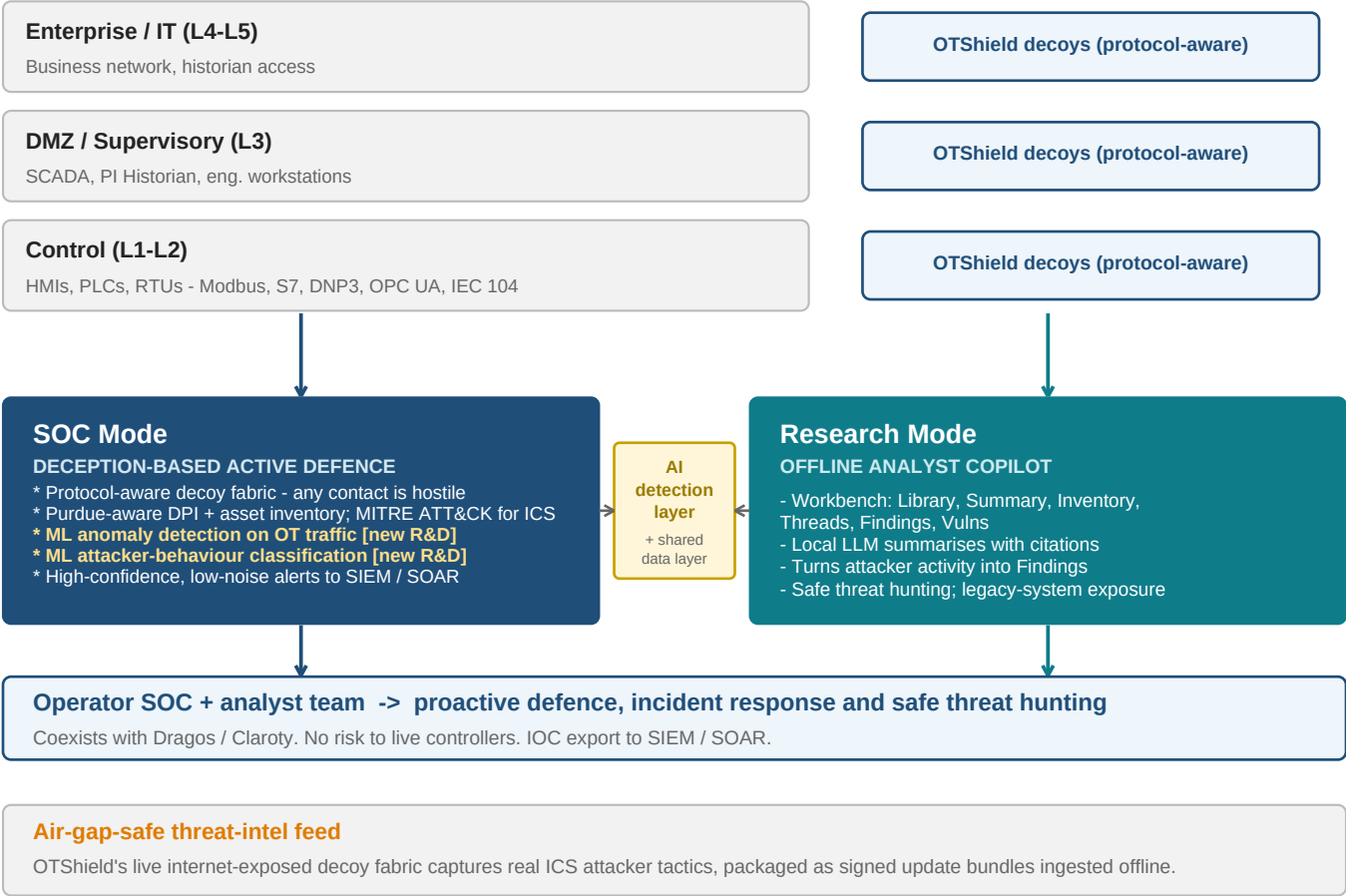
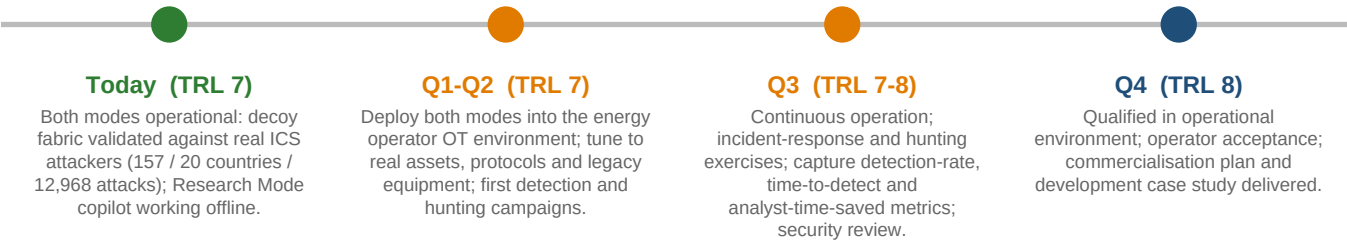


Table 1. Mapping to the four competition challenges

Challenge	Mode	How OTShield delivers it in this project
1. Proactive CNI defence & incident response	SOC Mode + AI	Decoy detections plus the new AI anomaly-detection layer; alerts to SIEM/SOAR; triage and response with the operator SOC team.
2. Detection & eviction of sophisticated actors	SOC Mode + AI	New AI attacker-behaviour classification surfaces protocol-aware intent (unauthorised writes, interlock overrides, remote discovery) auto-mapped to ATT&CK for ICS.
3. Threat hunting for CNI organisations	Research Mode	Copilot lets analysts query detections, manuals and asset data together and follow leads safely, with zero risk to live controllers.
4. Legacy system vulnerabilities in CNI	Both modes	SOC Mode inventories and risk-scores legacy assets; Research Mode copilot reasons over them to show how they could be reached and abused.

Figure 2. Readiness: TRL 7 today -> TRL 8 at project end



Why it meets the challenge

One integrated platform - SOC Mode for deception-based active defence, Research Mode for the analyst copilot - is proven, accepted and qualified inside a real UK energy operator, delivering all four challenge areas and ready to scale across the sector.